

MAT110

Differential Calculus and Coordinate Geometry

Lecture 10

Functions of Several Variables and Partial Derivatives

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1 Functions of Two and Three Variables

Reference: H. Anton, *Calculus*, Section 13.1.

Function of Two Variables

A function of two variables, x and y , is a rule that assigns a unique real number $f(x, y)$ to each point (x, y) in some set D in the xy -plane.

Function of Three Variables

A function f of three variables x , y , and z is a rule that assigns a unique real number $f(x, y, z)$ to each point (x, y, z) in some set D in three-dimensional space.

Example 01

Let

$$f(x, y) = \sqrt{y+1} + \ln(x^2 - y^2).$$

Find $f(e, 0)$, $f(x, 0)$, and $f(e, y)$.

Solution

$$\begin{aligned} f(e, 0) &= \sqrt{0+1} + \ln(e^2 - 0^2) \\ &= 1 + \ln e^2 = 1 + 2\ln e = 1 + 2 = 3, \\ f(x, 0) &= \sqrt{0+1} + \ln(x^2 - 0^2) = 1 + 2\ln x = h(x), \\ f(e, y) &= \sqrt{y+1} + \ln(e^2 - y^2) = g(y). \end{aligned}$$

For any fixed value of x , the function depends only on the one variable y . Also, for a fixed value of y , the function depends only on the one variable x , as illustrated by $h(x)$ and $g(y)$.

Example 02: Domain

Find the domain of

$$f(x, y) = xe^{-\sqrt{y+2}}.$$

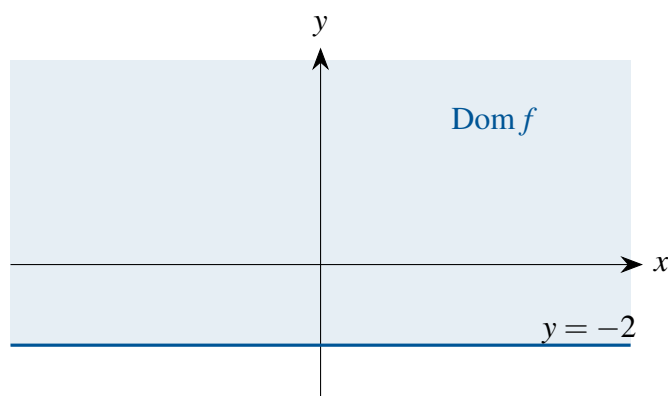
Solution

There is no value of x for which the function is undefined, so $x \in (-\infty, \infty)$. For y , we must have

$$y+2 \geq 0, \quad \text{that is,} \quad y \geq -2.$$

Thus, the domain consists of all points above or on the line $y = -2$ in the xy -plane:

$$\begin{aligned} \text{Dom } f &= \{(x, y) \in \mathbb{R}^2 : -\infty < x < \infty, -2 \leq y < \infty\} \\ &= \{(x, y) \in \mathbb{R}^2 : x \in (-\infty, \infty), y \in [-2, \infty)\}. \end{aligned}$$



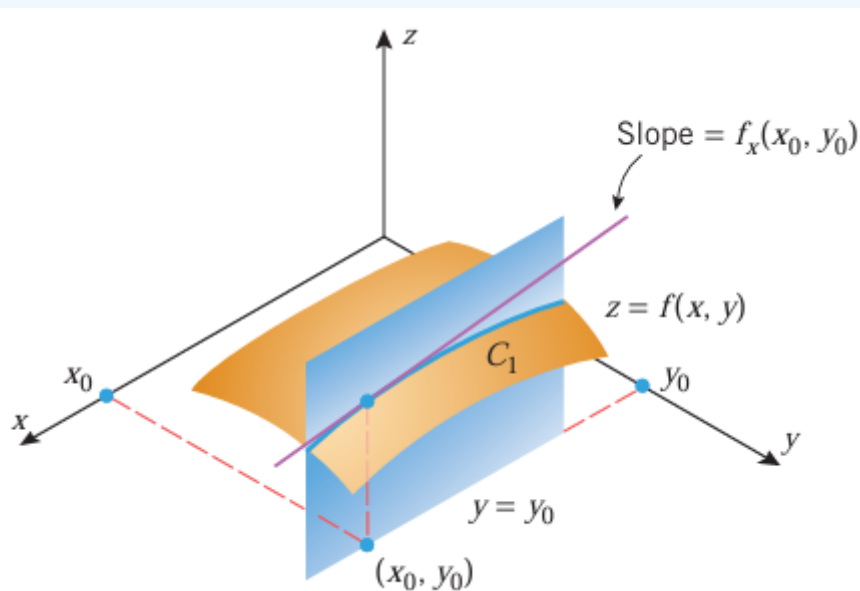
2 Partial Derivatives

Reference: H. Anton, *Calculus*, Section 13.3.

Partial Derivative with Respect to x (Definition 13.3.1)

If $z = f(x, y)$ and (x_0, y_0) is a point in the domain of f , then the partial derivative of f with respect to x at (x_0, y_0) is the derivative at x_0 of the function that results when $y = y_0$ is held fixed and x is allowed to vary. It is denoted by $f_x(x_0, y_0)$ and is given by

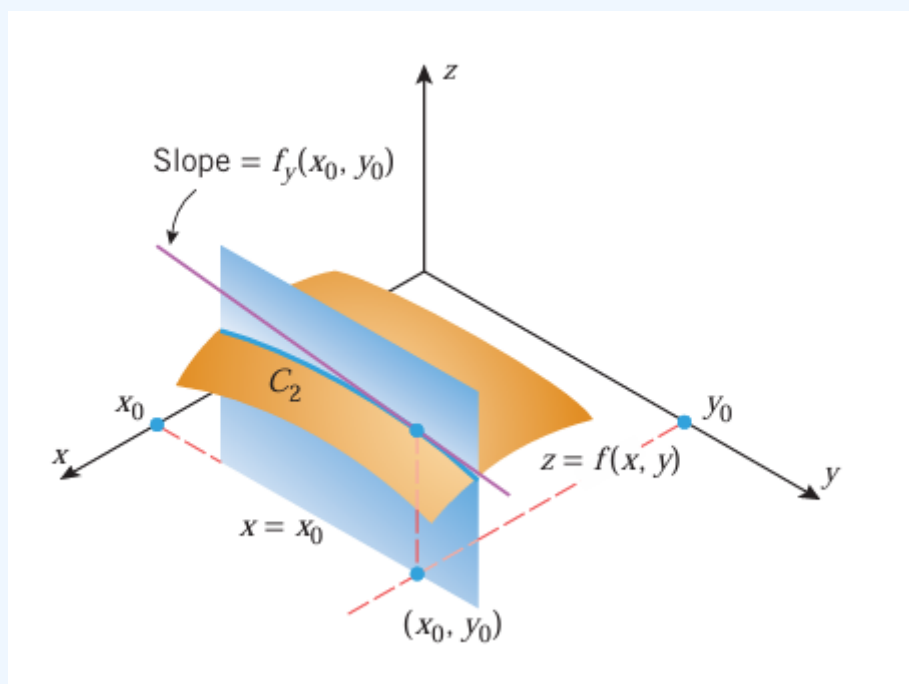
$$\begin{aligned} \frac{\partial f}{\partial x} = f_x(x_0, y_0) &= \left. \frac{d}{dx} f(x, y_0) \right|_{x=x_0} \\ &= \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h}. \end{aligned}$$



Partial Derivative with Respect to y

The partial derivative of f with respect to y at (x_0, y_0) is the derivative at y_0 of the function that results when $x = x_0$ is held fixed and y is allowed to vary. It is denoted by $f_y(x_0, y_0)$ and is given by

$$\begin{aligned}\frac{\partial f}{\partial y} &= f_y(x_0, y_0) = \left. \frac{d}{dy} f(x_0, y) \right|_{y=y_0} \\ &= \lim_{h \rightarrow 0} \frac{f(x_0, y_0 + h) - f(x_0, y_0)}{h}.\end{aligned}$$



2.1 Partial-Derivative Notation

If $z = f(x, y)$, then

$$f_x = \frac{\partial f}{\partial x} = \frac{\partial z}{\partial x}, \quad f_y = \frac{\partial f}{\partial y} = \frac{\partial z}{\partial y}.$$

At a particular point (x_0, y_0) ,

$$f_x(x_0, y_0) = \left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)} = \left. \frac{\partial z}{\partial x} \right|_{(x_0, y_0)},$$

and

$$f_y(x_0, y_0) = \left. \frac{\partial f}{\partial y} \right|_{(x_0, y_0)} = \left. \frac{\partial z}{\partial y} \right|_{(x_0, y_0)}.$$

Example 03

Find $\partial z / \partial x$ and $\partial z / \partial y$ for

(i) $z = f(x, y) = x^4 \sin(xy^3),$

(ii) $z = f(x, y) = x^2 + xy + y^3.$

Solution

For (i), treating y as a constant,

$$\begin{aligned}\frac{\partial z}{\partial x} &= \frac{\partial}{\partial x}(x^4 \sin(xy^3)) \\ &= x^4 \cos(xy^3)y^3 + 4x^3 \sin(xy^3),\end{aligned}$$

and, treating x as a constant,

$$\begin{aligned}\frac{\partial z}{\partial y} &= x^4 \frac{\partial}{\partial y} \sin(xy^3) \\ &= x^4 \cos(xy^3) 3xy^2 = 3x^5 y^2 \cos(xy^3).\end{aligned}$$

For (ii),

$$\begin{aligned}\frac{\partial z}{\partial x} &= \frac{\partial}{\partial x}(x^2 + xy + y^3) = 2x + y, \\ \frac{\partial z}{\partial y} &= \frac{\partial}{\partial y}(x^2 + xy + y^3) = x + 3y^2.\end{aligned}$$

Problem 01

Evaluate f_x and f_y for

- (a) $f(x, y) = x^2 + xy + y^4$,
- (b) $f(x, y) = \tan^{-1}(y/x)$,
- (c) $f(x, y) = 3x^3y^2 + 2y^2 + 4x$.

N.B. See Figure 13.3.1 from H. Anton to understand partial derivatives graphically.

2.2 Functions of More Than Two Variables

For a function $w = f(x, y, z)$ of three variables, the partial derivative $f_x(x, y, z)$ is calculated by holding y and z constant and differentiating with respect to x . For f_y , x and z are considered constants; for f_z , x and y are considered constants.

Example 04

Let

$$w = f(x, y, z) = x^3y^2z^4 + 2xy + z.$$

Find f_x , f_y , and f_z , and evaluate $f_z(-1, 1, 2)$.

Solution

$$\begin{aligned}\frac{\partial w}{\partial x} &= f_x(x, y, z) = 3x^2y^2z^4 + 2y, \\ \frac{\partial w}{\partial y} &= f_y(x, y, z) = 2x^3yz^4 + 2x, \\ \frac{\partial w}{\partial z} &= f_z(x, y, z) = 4x^3y^2z^3 + 1.\end{aligned}$$

Therefore,

$$f_z(-1, 1, 2) = 4(-1)^3(1)^2(2)^3 + 1 = -31.$$

3 Higher-Order Partial Derivatives

Suppose that f is a function of two variables, x and y ; that is, $f = f(x, y)$.

- Differentiating f twice with respect to x :

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial x} f_x = f_{xx}.$$

- Differentiating f twice with respect to y :

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} f_y = f_{yy}.$$

- Differentiating first with respect to x and then with respect to y :

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = f_{xy}.$$

- Differentiating first with respect to y and then with respect to x :

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = f_{yx}.$$

For example, a third-order partial derivative may be written as

$$\frac{\partial^3 f}{\partial x^2 \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) \right) = f_{yxx}.$$

Example 05

If

$$f(x, y, z) = \sin(3x + yz),$$

evaluate f_{xyz} and f_{xxyz} .

Solution

First,

$$\begin{aligned}
 f_x &= 3 \cos(3x + yz), \\
 f_{xy} &= \frac{\partial}{\partial y} (3 \cos(3x + yz)) = -3z \sin(3x + yz), \\
 f_{xyz} &= \frac{\partial}{\partial z} (-3z \sin(3x + yz)) \\
 &= -3z \cos(3x + yz)y - 3 \sin(3x + yz) \\
 &= -3yz \cos(3x + yz) - 3 \sin(3x + yz).
 \end{aligned}$$

For f_{xxyz} ,

$$\begin{aligned}
 f_x &= 3 \cos(3x + yz), \\
 f_{xx} &= -9 \sin(3x + yz), \\
 f_{xxy} &= -9z \cos(3x + yz), \\
 f_{xxyz} &= \frac{\partial}{\partial z} (-9z \cos(3x + yz)) \\
 &= -9[-zy \sin(3x + yz) + \cos(3x + yz)] \\
 &= 9yz \sin(3x + yz) - 9 \cos(3x + yz).
 \end{aligned}$$

Problem 02

Evaluate f_{xx} , f_{yy} , f_{xy} , f_{xxy} , f_{yyx} , and f_{yx} for the following functions:

- (a) $f(x, y) = \sin(x - e^y)$,
- (b) $f(x, y) = x^2 y^3 e^{xy^2}$,
- (c) $f(x, y) = e^{\sin(\sqrt{x^2 + y^2})} \tan(xy)$.

Practice Problems

From H. Anton:

Section 13.1: 1–4, 27,
Section 13.3: 1–10, 25–45.